

3.1.2 Section B: The living world

In this section, students are required to study [Ecosystems](#) (page 12), [Tropical rainforests](#) (page 12) and **one** from [Hot deserts](#) (page 13) or [Cold environments](#) (page 14).

3.1.2.1 Ecosystems

Key idea	Specification content
Ecosystems exist at a range of scales and involve the interaction between biotic and abiotic components.	An example of a small scale UK ecosystem to illustrate the concept of interrelationships within a natural system, an understanding of producers, consumers, decomposers, food chain, food web and nutrient cycling. The balance between components. The impact on the ecosystem of changing one component. An overview of the distribution and characteristics of large scale natural global ecosystems.

3.1.2.2 Tropical rainforests

Key idea	Specification content
Tropical rainforest ecosystems have a range of distinctive characteristics.	The physical characteristics of a tropical rainforest. The interdependence of climate, water, soils, plants, animals and people. How plants and animals adapt to the physical conditions. Issues related to biodiversity.
Deforestation has economic and environmental impacts.	Changing rates of deforestation. A case study of a tropical rainforest to illustrate: • causes of deforestation – subsistence and commercial farming, logging, road building, mineral extraction, energy development, settlement, population growth • impacts of deforestation – economic development, soil erosion, contribution to climate change.

Key idea	Specification content
Tropical rainforests need to be managed to be sustainable.	Value of tropical rainforests to people and the environment. Strategies used to manage the rainforest sustainably – selective logging and replanting, conservation and education, ecotourism and international agreements about the use of tropical hardwoods, debt reduction.

3.1.2.3 Hot deserts

Key idea	Specification content
Hot desert ecosystems have a range of distinctive characteristics.	The physical characteristics of a hot desert. The interdependence of climate, water, soils, plants, animals and people. How plants and animals adapt to the physical conditions. Issues related to biodiversity.
Development of hot desert environments creates opportunities and challenges.	A case study of a hot desert to illustrate: - development opportunities in hot desert environments: mineral extraction, energy, farming, tourism - challenges of developing hot desert environments: extreme temperatures, water supply, inaccessibility.
Areas on the fringe of hot deserts are at risk of desertification.	Causes of desertification – climate change, population growth, removal of fuel wood, overgrazing, over-cultivation and soil erosion. Strategies used to reduce the risk of desertification – water and soil management, tree planting and use of appropriate technology.

3.1.2.4 Cold environments

Key idea	Specification content
Cold environments (polar and tundra) have a range of distinctive characteristics.	The physical characteristics of a cold environment. The interdependence of climate, permafrost, soils, plants, animals and people. How plants and animals adapt to the physical conditions. Issues related to biodiversity.
Development of cold environments creates opportunities and challenges.	A case study of a cold environment to illustrate: • development opportunities in cold environments: mineral extraction, energy, fishing and tourism • challenges of developing cold environments: extreme temperature, inaccessibility, provision of buildings and infrastructure.
Cold environments are at risk from economic development.	The value of cold environments as wilderness areas and why these fragile environments should be protected. Strategies used to balance the needs of economic development and conservation in cold environments – use of technology, role of governments, international agreements and conservation groups.